

Research Statement

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Research Vision

My research aims to build **provenance-rich, fault-tolerant, and auditable AI infrastructures** for regulated financial systems. Modern document-AI pipelines handle billions in underwriting and compliance data, yet lack verifiable lineage, recovery guarantees, and compliance traceability. My goal is to unify **AI system reliability** and **distributed accountability** — ensuring every inference, document, and transaction in financial AI systems can be trusted, traced, and verified. **Background and Innovation**

As Founder of **CrocCRM** and Lead Engineer at **Pathway Catalyst**, I designed a live production system automating lender matching, underwriting, and compliance verification through AI-powered document orchestration. This practical foundation led to my **USPTO Provisional Patent #63/846,461**: “System and Method for Automated Recipient-Specific PDF Watermarking with CRM Integration.” The invention embeds provenance and audit credentials directly into document streams, bridging the gap between **AI explainability** and **systems accountability**. **Proposed Research Directions**

- Design **fault-injection and reproducibility frameworks** for distributed AI workflows.
- Develop **provenance graphs** that encode document lineage across compliance-critical systems.
- Extend **secure routing and attestation** principles from networked systems (e.g., SCION) into document-AI pipelines.
- Formally evaluate **energy-efficient and sustainable architectures** for large-scale, document-driven AI ecosystems. **Research Fit**

This work aligns closely with ongoing research at **MPI-SWS** (Druschel, Haeberlen), **EPFL** (Candea), **ETH Zürich** (Perrig), **VU Amsterdam** (Iosup, Lago), **Inria Paris** (Shapiro), **KTH** (Haller), and **Aalto University** (Ylä-Jääski). I am particularly interested in exploring **funded PhD or research positions** under these programs to formalize CrocCRM's applied engineering into verifiable, sustainable, and secure distributed AI systems. **Conclusion**

I believe the next generation of FinTech AI infrastructure must be built not only for speed and intelligence, but also for **provability, sustainability, and trust**. My mission is to bridge real-world financial automation with the academic rigor of dependable systems and document-AI research.